Endophyte literature check - rubric

Rubric — how to fill the judgement columns

Keep this open beside the workbook. Always record **what the text reports at the reading stage you're on** (abstract, or full text), not what you know about the taxon in general.

Dropdowns are suggestions — if the right answer isn't listed, **type it into the cell** (Excel warns but keeps it).

`text_reviewed` — set this when you finish a row; it decides what else you fill

value	when	then fill…
complete	the text reports its own primary fungus-from-plant data	the whole row
could not access the paper (full text needed)	a `full text` row where you couldn't get the PDF	nothing else — stop there
review or secondary compilation — no primary data of its own	a review, opinion, methods commentary, meta-analysis, or database paper whose text discusses endophytes but reports no fungus the authors themselves found	**Q1 = no · Q3 = review · Q4 (language)**. Leave Q2, Q5–Q11, the method ticks, and every interaction block **blank**. (Optionally note the review's topic in the summary column.)
not a fungus-in-plant study	wrong topic entirely — no plant, wrong organism, etc.	**Q1 = no**. Leave everything else blank.

A blank `text_reviewed` = you have not done that row yet. For a `complete` row, use the "not stated" / "unclear" option where the text is silent — **except the method columns** (those take `yes`/`no`/`not stated`; see the clarifications below) and unused interaction blocks, which stay blank.

Clarifications (added after the first reviewer's questions)

- **Abstract rows AND full-text rows — both.** All 20 papers in the shared block are full-text papers, so each has two rows: `K001a` (abstract — answer from the abstract text in that row only, don't open the PDF) then `K001b` (full text — open the PDF named `K001b.pdf`). 20 papers = 40 rows.
- **Method columns: fill every one on a `complete` row with `yes`, `no` or `not stated`.** `yes` = the text says that method was used. `no` = (full-text rows) you read the methods and that method was not used. `not stated` = you cannot tell from the text — this is the right answer on an **abstract** row unless the abstract itself says the method was used or not used. Blank always means you haven't done the row yet. The last column, `method not stated`, **fills itself in** (it shows `yes` only when all five methods are `not stated`, i.e. the text says nothing at all about how the fungus was shown to be inside the plant — common on abstract rows). Do not type in it.
- **Fungus names — record the lowest rank the authors name, exactly as written.**
  - Binomial given → the binomial (`Colletotrichum tropicale`). No variant/strain/isolate ID needed; if a strain code matters, mention it in `anything_you_were_unsure_about`.
  - Only a genus given (`Xylaria sp.`, `Xylaria`) → write it as the authors did.

  - No genus, only a higher rank (family, order, class, phylum — `Xylariaceae`, `Dothideomycetes`) ⟶ write that name; it still counts as a named taxon and gets a block.
  - Only sequence identifiers / OTU or ASV codes / "uncultured fungus" ⟶ **not** a named taxon: no block; describe them in `extra_pairs_or_NGS_community_summary` and answer Q5.
  - A name the authors give is recorded as-is even if you know it's outdated or misspelt.
- **Sampling location (Q7/Q8) — use only what the text itself says.** Read its own text, tables and figures/maps; do not open supplementary files, and do not follow a citation to another study.
  - **Coordinates are the most specific location there is, so always record them when the main text gives them** — in Q7c, even if it also names a place. Q7a is only a country the main text names, Q7b only a place name it gives; you never look anything up. Coordinates but no country or place name ⟶ Q7a = `not stated`, Q7b = `not stated`, Q7c = the coordinates as written, Q8 = `sampling location is explicitly stated`.
  - The main text says the sites are "as described in Smith et al. 2015" (or "see Supplement") and gives nothing itself ⟶ Q7a, Q7b and Q7c = `not stated`. **Q8 records the deferral**: `the main text says location details are in a supplement` or `... in another cited study` (we count these, so the study can report how often location was deferred). Name the cited study in `anything_you_were_unsure_about` if you like. A location that lives only in another paper is **not** "inferable" — `inferable` means the text itself names a place you can find on a map (a reserve, forest, island, city, region).
- **Blank cells.** A finished row is marked by `text_reviewed`, not by having no blanks. On a `complete` row: fill every Q1–Q11 cell and every method cell. For the interaction blocks, fill blocks 1…N where N is the number of named pairs (Q10, up to 5). **Unused blocks N+1…5 need nothing** — leave them empty. If you would rather make it explicit, type `N/A` in the `fungus` cell of an unused block; that is fine and is ignored when scoring either way. (`not stated` is different — it means a pair exists but the text is silent on that detail.) Inside a block you *do* fill, every one of the six cells gets an answer (`not stated` / `not reported` where the text is silent). The two wrap-up columns stay empty if you have nothing to add.
- **Where is this rubric?** It is the **Rubric tab** inside your workbook (a PDF copy is attached to the email if you want to print it). Tip: in Excel use View > New Window to see the Rubric tab beside the Review tab.

The paper-level questions

Q1 — does this text report a fungus coming from a plant? `yes` if the text isolated, detected, or observed at least one fungus *in or on* a living plant. `no` if it's the wrong organism, there is no plant host, or no fungus–plant association is reported at all. `unclear` as a last resort. - A natural-products paper that isolates compounds *from* an endophyte is still `yes` (its purpose goes in Q2). - A **review / compilation** whose text discusses endophytes but reports no fungus the authors found themselves ⟶ `no`; set `text_reviewed = review or secondary compilation` and follow that row of the table above. - If Q1 = `no`, `text_reviewed` cannot be `complete` — use one of the last two rows of the table above.

Q2 — the paper's main purpose (single choice; pick the one the abstract leads with) | value | e.g. | |—|—| | endophyte ecology / diversity / community | isolating/surveying endophytes of a host or region, community composition, biogeography, temporal change | | endophyte function: growth, nutrition, or stress tolerance | inoculation studies of biomass, yield, drought/salt/metal/heat tolerance, nutrient uptake, hormones | | endophyte function: disease protection / biocontrol | endophyte-mediated resistance to pathogens or pests; developing a biocontrol agent | | natural products / secondary metabolites / bioprospecting | screening endophyte extracts for bioactive / antimicrobial / anticancer compounds, enzymes | | plant pathology (the fungus studied as a pathogen) | the text is about disease caused by the fungus | | fungal taxonomy / systematics / new species | describing / revising taxa, phylogeny | | genomics / transcriptomics / methods development | a genome/transcriptome paper, or a new isolation/detection method | | other (type it) | anything else — type it |

Q3 — how were the fungi obtained in this text? (single choice) one or a few isolates (named) · culture survey (many isolates, named) · community metabarcoding/NGS (OTU/ASV table) · direct observation only (microscopy/histology, no isolation or sequencing) · **mixed methods (type which)** — after choosing this, type the combination, e.g. "mixed: culture + metabarcoding" · **review / synthesis / secondary compilation — no new fungi obtained here** (set `text_reviewed` to match — see the table above) · other (type it) · unclear. The method columns record the detail for the non-review cases.

Q4 — language of the text you are reading on this row. Answer for the reading stage. - English abstract of a paper written in French ⟶ `English` on the `abstract` row, `French` on the `full text` row. - **Bilingual abstract** (e.g. given in both English and Spanish) ⟶ `English + non-English` on the `abstract` row. - English abstract but the full text is another language ⟶ `English + non-English` on the `full text` row (and note the body language). Type the specific language after choosing `non-English (type which)`.

Q5 — any sequence-only (uncultured) taxa? `yes` if any reported fungi were detected **only by DNA sequencing, never cultured** (OTUs, ASVs, clone sequences).

Q6 — strain-level variation noted? `yes - noted` if the text says different strains/isolates of one species behaved differently. `only one strain / isolate studied` if there was nothing to compare. `no - not noted` otherwise.

Q7 — where was the plant SAMPLED? Three cells, all about the place the plant material was **collected**, using only what the MAIN TEXT you're reading says (on an abstract row, "the main text" is the abstract) — **you never need to look anything up**.

- **Q7a country** (dropdown, or type it) — the country where it was collected, as a standard English name. **This is the cell we score.** Several countries ⟶ `Brazil; Peru`. Use only a country the main text actually names. If the text gives only coordinates (no country name), or doesn't say where sampling was done, `not stated`.
- **Q7b site / locality** (free text) — state/province/region/reserve/site name as the main text gives it (don't go hunting). `not stated` if none given. Not scored.
- **Q7c coordinates** (free text) — if the main text gives latitude/longitude for the sampling site, type them **exactly as written** (e.g. `4°35'N, 74°04'W` or `4.59, -74.07`), even if it also names a place. `not stated` if none are given. Several sites: list up to three, separated by `;`. More than three: type `multiple sites (N) - see Table 1` (or wherever they are). Any of these counts as "coordinates given" for Q8.

Never put the authors' institution / affiliation country in any of these. Q7 is *only* the sampling location.

Q8 — what geographic information does the text contain? Answers both "how specific is it" and "where does the text keep it" in one question: - `sampling location is explicitly stated` — a named place or country, or coordinates, given directly in the main text - `sampling location is inferable from a named field site / region` — e.g. a named forest or reserve you can place on a map, without an explicit country/coordinates - `the main text says location details are in a supplement` / `... in another cited study` — the main text points elsewhere and gives nothing itself (e.g. "sites as described in Smith et al. 2015", "see Supplementary Table S1"). You do **not** open the supplement or the cited study; you only record that the main text points there. (If the main text gives a country directly but defers the exact site, that's still `explicitly stated` — pick whichever of the first two options matches what you actually got from the main text, and note the deferral in `anything_you_were_unsure_about` if you like.) - `no sampling location - only the authors' institutional country is given` — the text only tells you where the authors are - `no geographic information at all`

If Q7a = "not stated" and you can see the authors' affiliations, that is `no sampling location - only the authors' institutional country is given` — do **not** move the affiliation into Q7a.

Q9 — biome (WWF category). Only answer from what the text explicitly describes — a stated habitat, ecosystem, or biome term, or a description specific enough to place with confidence ("lowland tropical rainforest", "high-elevation páramo", "irrigated wheat field"). **Do not infer a biome from the host plant, the country, or a vague setting** — that guessing is exactly what this question exists to avoid; it would fold your own assumptions about climate into the "ground truth." If the text doesn't describe the growing environment, use `not stated`.

If a study spans more than one biome, list them comma-separated. Guidance for matching a *described* habitat to a WWF category: - **cultivated grassland** (sown pasture, hayfield, managed lawn) → `agriculture / cultivated`. Natural / semi-natural grassland, steppe, prairie → `temperate grassland / steppe / prairie`. - **forest** — pick by climate, not tree density: tropical/subtropical humid → `tropical / subtropical moist forest`; temperate, boreal, or a plantation → `temperate or boreal forest`; seasonally dry tropical → `tropical / subtropical dry forest`. - **estuary / salt marsh** (not mangrove) → `wetland / flooded grassland / bog` or `marine / coastal`. - greenhouse / pot experiment with no field component → `greenhouse / growth chamber`. - none fit → `other (type it)` and type it.

Why not Köppen–Geiger? KG (Kottek et al. 2006) is a more standardized climate system than WWF/Olson, but it classifies a **coordinate**, not what a text **says** — and using it here would mean the annotator has to go look up a KG map, which is exactly the "authors' terms might not match" problem KG was suggested to solve. Q9 stays a record of the text's *own* habitat description (or `not stated`) because that's something a reader can verify without guessing. Separately, for texts with coordinates (Q7 coordinates), we plan to compute an objective KG class programmatically after review — that covers texts with coordinates but no habitat description, the mirror of a text that describes habitat but gives no coordinates (which only Q9 catches). The two are complementary, not redundant.

Q10 — how many distinct fungus-host pairs? The number of distinct fungus × host-plant pairs the text **names** — write the true count, even if it is more than 5. Never a raw OTU/ASV count (put "10,412 ASVs" in the summary column instead). The sheet has room for 5 interaction blocks: if the text names more than 5, fill the 5 most prominent (the ones in the title/abstract, or the most abundant / most emphasised), put the rest in `extra_pairs_or_NGS_community_summary`, and still write the true count in Q10.

Q11 — was the surface-sterilisation checked? A standard methodological critique of culture-based endophyte studies: a surface-dwelling (epiphytic) fungus that survives the sterilisation step can be mistaken for a true endophyte once the tissue is cut open and plated. The only way to know the sterilisation actually worked is a **separate verification step**, done in addition to the sterilisation itself. Two things to tell apart: - **The sterilisation procedure itself** (e.g. "surface-sterilised by sequential immersion in 70% ethanol and 2% NaOCl") — this alone is not a check, it's just the prep. - **A verification / control step** — something that would show contamination *if* the sterilisation had failed: plating an aliquot of the **final rinse water** (should grow nothing), or pressing/imprinting the sterilised surface onto agar before cutting it open (should grow nothing). If the text reports one of these, that's `yes`.

value	use when…
`yes - a control was done`	the text reports a rinse-plating, tissue-imprint, or equivalent negative control, in addition to the sterilisation steps
`no - standard sterilisation, no check described`	the text describes the sterilisation protocol (solutions, timings) but reports no separate verification step

value	use when…
`not a culture-based study`	metabarcoding-only or microscopy-only — no isolation happened, so this QC step doesn't apply the same way
`cannot tell from the text`	genuinely ambiguous, or too little methods detail to judge

Worked contrast: "leaves were surface-sterilised (70% EtOH, 30s; 2% NaOCl, 2min; rinsed 3x sterile water) before plating on PDA" ⟶ `no - standard sterilisation, no check described` (no verification mentioned). Add "…; sterility of the technique was confirmed by plating 100 µL of the final rinse water, which showed no microbial growth" ⟶ `yes - a control was done`.

(Whether the fungus was also confirmed *inside* the tissue by microscopy or resynthesis is a different question, already captured by the "microscopy in tissue" / "resynthesis / re-inoculation" method columns — Q11 is specifically about ruling out surface contamination.)

Method columns (the purple block) — fill every one: `yes` / `no` / `not stated`

column header	`yes` when the text… (`no` = methods read, not used; `not stated` = can't tell)
`culture from sterilised tissue`	surface-sterilised tissue, plated it, grew fungi out
`microscopy in tissue`	showed hyphae *inside* tissue by staining / sectioning / confocal / TEM (covers "histology")
`direct sequencing from tissue`	sequenced fungal DNA straight from tissue — metabarcoding, clone libraries (no culturing step)
`isolate ID by sequencing`	Sanger / ITS-sequenced the *cultured isolates* to identify them
`resynthesis / re-inoculation`	re-inoculated a sterile plant with an isolate and re-isolated it
`method not stated`	fills itself in: `yes` when all five methods above are `not stated` — do not type in it

Whether the sterilisation was *checked with a control* goes in Q11, not here. On a `complete` row no method cell is left blank.

Interaction blocks — one per DISTINCT fungus × host-plant pair the text NAMES

Each block: `fungus` · `host_plant` · `tissue` · `fungal_lifestyle` · `effect_on_host` · `evidence_basis`.

- Fill blocks only for taxa the text **names**. For a metabarcoding community, name the dominant few and summarise the rest in `extra_pairs_or_NGS_community_summary`.
- **`fungus` / `host_plant` — copy the name exactly as the authors write it**, genus + species where they give it. If you happen to know the taxon has since been renamed, **still record the authors' name, not the current one.** Synonym resolution to current taxonomy is done programmatically afterwards; changing names by hand would corrupt that step. (Spelling fixes for obvious typos are fine.)

- **tissue** — if one fungus + host was found in more than one tissue, put them all in the one cell, comma-separated: `leaf, root`. Don't make a separate block per tissue. If the text only says "shoots" / "aerial parts" without separating the tissues, choose `aerial parts / shoots (not separated)` — do not guess leaf or stem.

`fungal_lifestyle` — how the fungus makes its living (trophic mode)

Values follow **FungalTraits `primary_lifestyle`** (Põlme et al. 2020, *Fungal Diversity* 105:1–16). FungalTraits' full plant-relevant list is: *foliar endophyte, root endophyte, root endophyte-dark septate, plant pathogen, plant pathogen-biotroph/-hemibiotroph/ -necrotroph, arbuscular mycorrhizal, ectomycorrhizal, ericoid mycorrhizal, orchid mycorrhizal, wood saprotroph, litter saprotroph, soil saprotroph, plant/unspecified saprotroph, nectar/tap saprotroph, epiphyte, sooty mould, mycoparasite, lichenised, lichen parasite.* We collapse the three endophyte types into one "endophyte" value (the `tissue` column already records which tissue — leaf, root, stem, or otherwise — so stem tissue is just as much "endophyte" as leaf or root; endophyte does not imply saprotroph); the four mycorrhizal types are kept **separate** (arbuscular / ecto / ericoid / orchid).

value	use when the fungus is…
endophyte (any plant tissue)	living inside healthy tissue, no disease — the default for this literature. Any tissue counts (leaf, root, stem, …); the `tissue` column already records which, so residing in stem tissue does **not** imply it must be a saprotroph
latent pathogen / hemibiotroph	symptomless now but the text shows/states it turns pathogenic (later, under stress, on other hosts)
plant pathogen	causing disease in the studied interaction
wood saprotroph	decaying woody tissue
litter saprotroph	on fallen leaves / needle litter
soil saprotroph	free-living in soil (recovered from the root zone but not the plant)
unspecified / other saprotroph	dead/senescent plant tissue, type not specified
arbuscular mycorrhizal	an AM (Glomeromycota) root symbiont
ectomycorrhizal	an ECM root symbiont (sheath / Hartig net; e.g. *Russula*, *Cortinarius*, *Tuber*)
ericoid mycorrhizal	an ericoid symbiont of Ericaceae hair roots (e.g. *Rhizoscyphus ericae*)
orchid mycorrhizal	a symbiont of orchid roots / protocorms (e.g. *Tulasnella*, *Ceratobasidium*)
mycoparasite / antagonist of other microbes	attacking / suppressing other fungi or microbes
epiphyte (surface only)	on the plant surface, not inside
lichenised	part of a lichen
context-dependent	the text says the lifestyle itself shifts with host/environment/strain
not clear from the text	

`effect_on_host` — the net outcome for THIS host plant, as the text reports it

value	use when the text…
harmful to the plant	reports disease, symptoms, reduced growth or fitness
beneficial to the plant	reports a **measured or clearly stated** benefit — growth, stress tolerance, defence, nutrition
neutral / commensal	**tested the plant's response** (growth, fitness, stress, defence) **and found no net effect**
no visible symptoms; benefit or harm not tested	says the plant looked healthy / "symptomless", but did not test for benefit or harm
not reported	says **nothing** about effect — just isolated / detected the fungus
context-dependent	outcome varies by host genotype / environment / strain
unclear	you genuinely cannot tell

Why lifestyle and effect are separate columns: the same fungus can appear as a `plant pathogen` lifestyle with `no visible symptoms` effect (a latent infection), or an `endophyte` lifestyle that turns out `harmful`. Keeping them apart is the whole point of the study — "endophytism is a location, not a fixed function."

The three "not obviously harmful" effects differ by **how much the text checked**:

	mentioned symptoms?	measured the plant's response?
not reported	no	no — "endophytic fungi were isolated from leaves"
no visible symptoms, not tested	yes ("symptomless endophyte")	no
neutral / commensal	usually	**yes** — and the effect was ~zero

`evidence_basis` — how the effect on the host you recorded was established

If `effect_on_host` is `not reported`, record how the **lifestyle** was established instead. (One rule, so everyone answers the same way.)

value	use when…
experimentally tested in this study	inoculation / re-synthesis / knockout / a controlled assay
observed / measured in this study (no manipulation)	co-occurrence, community stats, expression, field observation of symptoms
inferred from what the fungus usually does	assigned from the genus/species name or a database, not data in this text
asserted with no support	stated in passing, no evidence
not stated	

Worked examples

1 — abstract, nothing tested. *"Endophytic fungi were isolated from surface-sterilised leaves of Espeletia in the páramo of Chingaza, Colombia; dominant taxa Xylaria sp. and Nigrospora sphaerica."* → Q1 yes · Q2 endophyte ecology · Q3 culture survey · Q7a Colombia · Q7b Chingaza páramo · Q7c not stated · Q8 explicitly stated · Q9 montane grassland / alpine (the abstract's own word "páramo" is explicit enough) · Q10 2 · Q11 cannot tell from the text (the abstract does not say whether sterilisation was checked) · methods: `culture from sterilised tissue` yes, the other four `not stated` · 2 blocks, each: tissue leaf / lifestyle `endophyte` / effect `not reported` / evidence `inferred from what the fungus usually does`. (If the abstract had said only "Espeletia" with no place, Q7a/Q7b/Q7c would all be `not stated`, Q8 `no geographic information at all`, and Q9 `not stated` too — we never fill in a place, or a habitat, from what we know about the plant.)

2 — full text, benefit tested. Wheat inoculated with *Epichloë*, +22% biomass under drought. → Q3 one/few isolates · `resynthesis / re-inoculation` yes · Q11 as the text reports · block 1: *Epichloë* sp. / *Triticum aestivum* / whole plant / lifestyle `endophyte` / effect `beneficial to the plant` / evidence `experimentally tested in this study`.

3 — latent pathogen. *"Diplodia sapinea recovered asymptomatically from Pinus shoots; water-stressed inoculated seedlings developed shoot blight."* → block 1: *Diplodia sapinea* / *Pinus* sp. / stem-wood-bark / lifestyle `latent pathogen / hemibiotroph` / effect `harmful to the plant` (under water stress) / evidence `experimentally tested in this study`. Note the stress condition in the summary column.

4 — metabarcoding. *"ITS metabarcoding of Quercus robur leaves recovered 142 fungal OTUs (38 genera); most abundant Cladosporium, Aureobasidium."* → Q3 community metabarcoding · Q5 yes · Q11 not a culture-based study · `direct sequencing from tissue` yes · block 1: *Cladosporium* / *Quercus robur* / leaf / lifestyle `endophyte` / effect `not reported` / evidence `inferred...` · block 2: *Aureobasidium* / … · summary: *"142 OTUs / 38 genera from Q. robur leaves; asymptomatic foliar endophyte community; no effect tested."*